

Table 7-1. Quantities printed to file SOLUTION.RES

Quantity	Dimensionless to k	Dimensionless to d
Water depth	kd	$d/d = 1$
Wave length	$k\lambda = 2\pi$	λ/d
Wave height	kH	H/d
Wave period	$\tau\sqrt{gk}$	$\tau\sqrt{g/d}$
Wave speed	$c\sqrt{k/g}$	$c/\sqrt{gd}$
Eulerian current	$\bar{u}_1\sqrt{k/g}$	$\bar{u}_1/\sqrt{gd}$
Stokes current	$\bar{u}_2\sqrt{k/g}$	$\bar{u}_2/\sqrt{gd}$
Mean fluid speed	$\bar{U}\sqrt{k/g}$	$\bar{U}/\sqrt{gd}$
Wave volume flux, $q = \bar{U}d - Q$	$q\sqrt{k^3/g}$	$q/\sqrt{gd^3}$
Bernoulli constant, $r = R - gd$	rk/g	r/gd
Volume flux	$Q\sqrt{k^3/g}$	$Q/\sqrt{gd^3}$
Bernoulli constant	Rk/g	R/gd
Momentum flux	Sk^2/pg	S/pgd^2
Impulse	$I\sqrt{k^3}/(\rho\sqrt{g})$	$I/(\rho\sqrt{gd^3})$
Kinetic energy	Tk^2/pg	T/pgd^2
Potential energy	Vk^2/pg	V/pgd^2
Mean square of bed velocity	$u_\beta^2 k/g$	u_β^2/gd
Radiation stress	$S_{xx}k^2/pg$	S_{xx}/pgd^2
Wave power	$Fk^{5/2}/(pg^{3/2})$	$F/(pg^{3/2}d^{5/2})$
Fourier coefficients	B_j	E_j